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PROJECT EXPERIENCE

Shorelife: California Beach Health Forecasting (iOS app, live on App Store) Apr 2026 – Present

- Engineered a soft ensemble of 10 undersampled XGBoost models for beach-bacteria forecasting, 0.80 AUCPR
- Beat the persistence baseline on spatial hold-outs, 0.96 vs 0.82 AUCPR across 15 held-out beach folds
- Engineered ETL pipelines merging marine microbiology, hydrology, stormwater, and weather sources
- Deployed a FastAPI service through Render; daily retraining via GitHub Actions, 296 runs since Apr 2026
- Open-sourced the ML pipeline and published a model card under AGPL for outside review

Senior Capstone, industry partner Prism Data: Transaction Intelligence and Credit Risk Sep 2025 – Mar 2026

- Built a transformer from scratch in PyTorch to categorize bank transactions, achieving 95% accuracy and F1
- Soft-ensembled 10 XGBoost models on undersampled data into a bagged-boosted model reaching 0.86 ROC-AUC
- Chose undersampling after synthetic data failed to carry the essence of the real distribution
- Built a local LLM agent workflow testing feature hypotheses privately, cutting manual feature work by 50%

XOXO: On-Device Generative Avatars, Apple Image Playground (iOS, v1.0 in App Review) Aug 2026 – Present

- Built a SwiftUI and CloudKit app where partners share a mood as a face generated fully on-device
- Designed a prompt-evaluation harness scoring mood differentiation against identity drift via Vision
- Found a word cap halved duplicate outputs and presence-conditional phrasing fixed 3 of 3 hallucinations
- Automated rounds on 4 iPhones and 2 iOS releases via XCUITest; identity held on 16 of 16 seed photos

Chef_LLM: Digitizing Taste (in progress, github.com/kylechoi101/chef_LLM) Jul 2026 – Present

- Hypothesis: recipes exist to be reproduced; a recipe is a program, cooking is execution, taste is the output
- Built ETL and validation over 231,637 recipes and 1.07M ratings, clustering into 211K recipe families
- Invalidated the quality heuristic: the largest lint flag hit 19% of the corpus yet matched passing ratings

Cross Market Lag Research (open source: github.com/kylechoi101/cross-market-lag-research) May 2026

- Hypothesis: one global market influences a second hours apart, so investors can benefit off that lag
- Built an ingestion and backtest pipeline across 13 markets, 156 directed pairs, and 588 coupling analyses
- Validated with a permutation null and rejected candidates via DSR, PBO, White's Reality Check, cost stress
- Found 60 coupled market pairs but only 1–22% transmission to the tradable ETF; published the negative result

LLM Evaluation Research: [Time-Series Dynamics in Baduk](#) (Go) May 2026 – Jun 2026

- Modeled Go dynamics as a time series, reducing it to a 5-variable conditional logit over 361 board points
- Measured coupling: black's last move explains 44% of white's reply, within three points 56% of the time
- Documented forced attention: naming the LLM a well-documented scholar pulls output toward ground truth

MLB Swing Zone Visualization Apr 2025 – Jun 2025

- Visualized swing decisions for 100+ MLB batters expanding their swing zone by 35% as the strike count rises
- Built an interactive 3D web visualization per selected hitters, presented to 30+ non-technical audience

WORK EXPERIENCE

UCSD, San Diego California – Research Assistant Jun 2024 – Feb 2025

- Recruited into a modular neutrino detector group; picked up the lab's ML stack alongside PhD researchers
- Distributed detector processing jobs and worked hands-on with SDSC compute and FPGA-based readout

Army, Republic of Korea – Sergeant Feb 2021 – Aug 2022

- Commanded a unit of 25+; oversaw artillery and coordinated with first responders during the Covid-19 pandemic

EDUCATION

University of California San Diego — B.S. Data Science, Jun 2026

- Major GPA:** 3.8 / 4.0; Cumulative GPA: 3.7 / 4.0
- Coursework:** Time-Series Analysis, Probabilistic Modeling, Representation Learning, Financial Mathematics
- Awards:** HDSI Best Visualization Project, Rady Advanced Fundamental Equity Valuation Tournament

Chartered Financial Analyst Exam, CFA® Institute — Passed Levels I and II; Level III result pending

SKILLS

- Languages:** Python, SQL, Swift, JavaScript, R
- ML:** PyTorch, XGBoost, scikit-learn, transformers, ensembling and calibration, time-series modeling, NLP
- LLM / AI engineering:** eval harness design, multi-agent workflows, local LLMs, Apple Foundation Models
- Infrastructure:** FastAPI, GitHub Actions CI/CD, Render, AWS, Firebase, Docker, D3
- Spoken:** Korean, English, Mandarin Chinese